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Gilbert's syndrome and the risk of death: a population-based cohort study

Laura J. Horsfall, Irwin Nazareth, Stephen P. Pereira, Irene Petersen

Author names and affiliation:

Laura J. Horsfall, PhD [1]

Irwin Nazareth, PhD, FRCGP [1]

Stephen P. Pereira, PhD, FRCP [2]

Irene Petersen, PhD [1]

1. Research Department of Primary Care and Population Health, Institute of Epidemiology and Health Care, University College London, United Kingdom
2. Institute of Liver and Digestive Health, University College London, United Kingdom

Contact information:

Laura Horsfall

Research Department of Primary Care & Population Health

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Faculty of Biomedical Sciences

University College London Medical School

Royal Free Campus

Rowland Hill Street

London NW3 2PF

Tel: +44 (0) 207 794 0500

Fax: +44 (0) 207 794 1224

email: laura.horsfall@ucl.ac.uk

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Abstract

Background & aims: Gilbert's syndrome is a common familial hyperbilirubinemia that may reduce the risk of various age-related diseases due to the antioxidant properties of bilirubin. We conducted a large cohort study using The Health Improvement Network (THIN) primary care database and compared all-cause mortality rates in those with and without Gilbert's syndrome.

Methods: Mortality rates in patients with a diagnosis of Gilbert's syndrome and raised bilirubin level (n=4,266) were compared to those of patients with similar characteristics but with normal bilirubin levels (n=21,968). Multivariable Poisson regression was also used to estimate adjusted mortality rate ratios (MRRs).

Results: During the 35,0000 PYs of follow-up across the Gilbert's and comparison cohorts there were 1174 deaths. Mortality rates were 24/10,000 PYs in the Gilbert's cohort versus 50/10,000 PYs in the comparison cohort. Mortality rates were around half in patients with Gilbert's syndrome after accounting for sociodemographics and general health indicators (adjusted MRR: 0.5 [95%CI; 0.4-0.7; p<0.001]).

Conclusions: Mortality rates observed for people with Gilbert's syndrome in the general population are almost half those of people without evidence of Gilbert's syndrome.

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Keywords (MeSH):

Gilbert Disease; Bilirubin uridine-diphosphoglucuronosyl transferase 1A1;Bilirubin; mortality; Database; Cohort.

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Introduction

Gilbert's syndrome (MIM#143500) is a common form of unconjugated hyperbilirubinemia caused by an inherited deficiency in the bilirubin-conjugating enzyme UDP-glucuronosyltransferase 1-1 (UGT1A1) [1]. The condition is highly prevalent ranging from 3-12% of the population depending on the mode of diagnosis [1-3]. Intermittent jaundice is the only clinically recognised symptom, which can be triggered by infections, certain drugs and situations associated with low caloric intake such as gastrointestinal illness, surgery and fasting [4-7].

Gilbert's syndrome is historically regarded as a benign trait and clinical recognition is often incidental to routine laboratory investigations [8-10]. However, there has been renewed interest in the condition following the results from more recent genetic association studies. For instance, homozygosity for the gene promoter variant termed UGT1A1*28, which underlies the condition in Europeans [1], has been associated with an increased risk of developing gallstones [11-13] and also a predisposition to certain adverse reactions to drugs requiring UGT1A1-mediated glucuronidation for elimination [14, 15]. Raised bilirubin may protect against a range of age-related diseases due to the potent antioxidant/anti-inflammatory properties [16, 17]. Indeed, markers of oxidative stress appear reduced in people with Gilbert's syndrome[18] and much lower odds of cardiovascular disease (CVD) have been reported in European and Chinese people with *UGT1A1*

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variation underlying the condition [17, 19, 20].

There have been no large-scale studies of Gilbert's syndrome in the general population and the aim of this study was to use The Health Improvement Network (THIN), a United Kingdom (UK) primary care database, to estimate the risk of death from any cause compared to patients without evidence of Gilbert's syndrome.

Patients and methods

Data source

Most primary care physicians (general practitioners -GPs) in the UK record patient data electronically and some practices have opted to provide these data for clinical and epidemiological research. THIN primary care database provides anonymized records on over nine million patients registered with over 500 general practices that use the VISION software to record details of consultations (http://csdmruk.cegedim.com/). Information on diagnoses, symptoms, and referrals to secondary care are electronically recorded as Read codes, a hierarchical coding system used in UK primary care [21]. Information on prescriptions and variables such as height, weight, blood pressure, smoking status and laboratory test results are also recorded. The database provides quintiles of the Townsend score measure of social deprivation, which is a composite measure of owner-occupation, overcrowding, car ownership, and unemployment levels derived from UK census data and linked to the patient's postal code. THIN data is broadly representative of the UK general practice population in terms of

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demographics and consultation behaviour [22, 23]. Clinical diagnoses recorded by GPs electronically have recently been shown to be accurate compared with other reliable sources [23, 24].

A cohort design was used to compare mortality rates in patients with a diagnosis of Gilbert's syndrome to those without evidence of Gilbert's syndrome. All patients with both an abnormal bilirubin level (>17 μmol/L) and a Read code diagnosis of Gilbert's syndrome (code C374200) were extracted from the database. Within each general practice, a comparison cohort with normal bilirubin levels (≤17 μmol/L) was extracted using stratified sampling to ensure similar characteristics in terms of age, sex and year of bilirubin test. Up to six comparator patients were selected per patient with Gilbert's syndrome. Patients entered the cohort at the earliest date of either a) the patient registering with the general practice or b) the date when the practice was deemed to record mortality at a rate similar to that of the United Kingdom given the age and gender structure of the practice (described elsewhere as the date of acceptable mortality recording) [25]. Patients exited the cohort at the end of the study period (December 31st 2010), when death was recorded or when they transferred to a different practice.

Outcomes

The rates of all-cause mortality were compared in the two cohorts. Death was ascertained by the presence of a date of death in the patient records.

Covariates

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General health indicators previously associated with bilirubin levels were selected as covariates as these may influence a diagnosis of Gilbert's syndrome. These included smoking status, body mass index (BMI), blood pressure, and the Townsend score of social deprivation. Alanine aminotransferase (ALT) and alkaline phosphatase (ALP) are almost always recorded at the same time and bilirubin and were also extracted to account for any differences in liver function. Where multiple values had been recorded, that taken closest to the bilirubin test date was selected. Age, gender and time period were also included to account for any imbalances in the stratified sampling. Bilirubin levels and/or *UGT1A1* variation has been associated with a range of conditions including diabetes, mental illness, cardiovascular and respiratory disease, and adjustments for these co-morbidities were made as a supportive analysis to examine the extent to which these conditions may play a role on the causal pathway with mortality. Read code lists for these conditions were developed using a previously reported method [26]. Drug prescriptions for insulin and psychotropic drugs were also used to define cases of diabetes and mental illness respectively.

Statistical analyses

Multivariable Poisson regression was used to estimate the mortality rate ratios (MRRs) for mortality in the Gilbert's syndrome cohort versus the comparison cohort. Poisson regression is appropriate for these longitudinal data as the event of interest (death) is rare. A complete-case analysis was done with continuous covariates entered as such into the model. We tested for interactions between gender and Gilbert's syndrome and smoking and

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Gilbert's syndrome by comparing nested models with and without interaction terms using the log-likelihood ratio test. To account for data clustering within GP practice, we used a multilevel Poisson regression model with the practice identifier entered as a random effect. All p-values were two-tailed and a value of less than 0.05 was considered statistically significant. All analyses were performed using Stata version 12.

The THIN Scheme was approved by the National Health Service South-East Multi-centre Research Ethics Committee (MREC) and the present study was approved by the Cambridgeshire MREC.

Results

The characteristics of the Gilbert's cohort (n=4,266) and the comparison cohort (n=21,968) were very similar except for a lower proportion of smokers and lower livels of co-morbidity at cohort entry in the Gilbert's cohort (Table 1). Over a total of 35,0000 PYs of follow-up there were 1174 deaths. After accounting for potential confounding factors including smoking and social deprivation, the mortality rates in people with Gilbert's syndrome were almost half those of the comparison cohort (Table 2). Including potential mediators of the association with mortality (diabetes, respiratory, and cardiovascular disease) in the regression model only slightly attenuated the association with mortality causing the adjusted MRRs for the Gilbert's cohort versus the comparison cohort to reduce in magnitude from 0.53 to 0.57. Including a history mental illness had no impact on the associations. Adjusting for smoking intensity (number of cigarettes smoked per day) also had no

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meaningful impact on the associations. There was a suggestion of an interaction with smoking status, with Gilbert's syndrome associated with 60% lower rates of mortality when both cohorts were restricted to current or ex-smokers versus 40% lower mortality rates for the Gilbert's cohort when restricted to non-smokers, but this interaction was not statistically significant (p=0.12).

Discussion

This study is the largest population-based analysis of diagnosed Gilbert's syndrome. Those with a diagnosis tend to have a much lower mortality rates compared to patients without evidence of the condition. The 50% lower mortality rates observed in patients with Gilbert's syndrome could reflect the systemic antioxidant/anti-inflammatory properties of serum bilirubin [16, 27-31]. The Framingham offspring Cohort study reported that CVD rates were 64% lower (95%CI; 26-82%;p=0.005) in patients homozygous for UGT1A1*28 compared with the other genotypes [17]. Mortality rates in a cohort of hemodialysis patients were 75% lower (95%CI; 21-98%; p=0.019) for UGT1A1*28 homozygotes relative to other genotypes over a 12-year period [19]. However, this relationship has not been replicated across all studies [32].

While it is plausible that the lower mortality rates in people with Gilbert's syndrome reflects protection against CVD, the major cause of death in the UK, it is also possible that protection from other diseases related to oxidative stress and inflammation also contribute to the relationship. Adjusting for conditions potentially on the causal pathway (diabetes, respiratory or

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cardiovascular disease) only had a modest effect on the associations. Moderately raised bilirubin may have other protective properties that explain the association with mortality but which were not accounted for in our study. For instance, there is some evidence from early studies that raised bilirubin may help protect against infection [33]. In addition, primary cause of death can be difficult to determine and is not always documented in the patient records using Read codes. Thus it might just be that we did not capture and account for all deaths due to diabetes, respiratory or cardiovascular disease in the analysis.

The major strength of this study is the sample size allowing fairly precise estimates on the relative mortality rates for patients with and without evidence of Gilbert's syndrome. However, GPs record information for the purpose of patient management and some variables useful for researchers but not directly relevant for patient care, have high levels of missing data. For instance, ethnicity may have also influenced diagnosis of Gilbert's syndrome and be unbalanced in the two cohorts but is not well recorded in primary care. However, the UK is over 92% white European and this is unlikely to be a strong source of confounding. There are no established clinical guidelines for diagnosing Gilbert's syndrome in the UK and therefore the mode of diagnosis in the primary care setting is unclear. Before commencing the study, we conducted some preliminary work to explore the general mode of diagnosis by examining all medical records immediately prior to diagnosis in a random subset of 100 cases. Just four individuals were recorded as presenting with jaundice. The remainder appeared to be diagnosed following routine bilirubin

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tests and only 10 cases were recorded as being referred to secondary care shortly before the diagnosis was recorded. Thus is seems likely that most diagnoses are made incidentally using exclusion criteria in the primary care setting. However, secondary care referrals are not always recorded in primary care records and other observations may have been entered as free text which was not available for analysis. In addition, confirmatory genetic testing for Gilbert's syndrome is not routine in the UK. Thus there is a possibility that some cases of Gilbert's syndrome are miss-diagnoses in the primary care setting. This could have diluted the results if a large number of non-cases were included in the Gilbert's cohort. There are also limitations on the generalizability of the findings. For instance, the results cannot be generalised to all patients with Gilbert's syndrome because most are undiagnosed and the group we have examined may be a more severe subset with additional genetic variation of *UGT1A1* or other enzymes associated with bilirubin elimination. The cohorts were restricted to patients with bilirubin tests in their medical history and although bilirubin testing is common in the UK, they may still represent a less healthy sub-population.

Patients with a record of Gilbert's syndrome have lower mortality from any cause, which may represent the beneficial properties of moderately raised bilirubin. The suggestion of an interaction with smoking status is interesting and warrants further research. Whether the relationship between Gilbert's syndrome and lower mortality is causal remains to be proven.

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References

[1] Bosma PJ, Chowdhury JR, Bakker C*, et al.* The genetic basis of the reduced expression of bilirubin UDP-glucuronosyltransferase 1 in Gilbert's syndrome. *N Engl J Med*. 1995; **333**: 1171-5.

[2] Owens D, Evans J. Population studies on Gilbert's syndrome. *J Med Genet*. 1975; **12**: 152-6.

[3] Buyukasik Y, Akman U, Buyukasik NS*, et al.* Evidence for higher red blood cell mass in persons with unconjugated hyperbilirubinemia and Gilbert's syndrome. *Am J Med Sci*. 2008; **335**: 115-9.

[4] Cleary KJ, White PD. Gilbert's and chronic fatigue syndromes in men. *Lancet*. 1993; **341**: 842.

[5] Ashraf W, van Someren N, Quigley EM, Saboor SA, Farrow LJ. Gilbert's syndrome and Ramadan: exacerbation of unconjugated hyperbilirubinemia by religious fasting. *J Clin Gastroenterol*. 1994; **19**: 122-4.

[6] Powell LW, Hemingway E, Billing BH, Sherlock S. Idiopathic unconjugated hyperbilirubinemia (Gilbert's syndrome). A study of 42 families. *N Engl J Med*. 1967; **277**: 1108-12.

[7] Bakhotmah MA, Gasem AA, Bairotee B. Asymptomatic unconjugated hyperbilirubinemia (Gilbert syndrome) among Saudis in Jeddah. *Ann Saudi*

This article is protected by copyright. All rights reserved.

Accepted Article

Med. 1995; **15**: 422-3.

[8] Teich N, Lehmann I, Rosendahl J, Troltzsch M, Mossner J, Schiefke I. The inverse starving test is not a suitable provocation test for Gilbert's syndrome. *BMC Res Notes*. 2008; **1**: 35.

[9] Metreau JM, Yvart J, Dhumeaux D, Berthelot P. Role of bilirubin overproduction in revealing Gilbert's syndrome: is dyserythropoiesis an important factor? *Gut*. 1978; **19**: 838-43.

[10] Olsson R, Stigendal L. Clinical experience with isolated hyperbilirubinemia. *Scand J Gastroenterol*. 1989; **24**: 617-22.

[11] Buch S, Schafmayer C, Volzke H*, et al.* Loci from a genome-wide analysis of bilirubin levels are associated with gallstone risk and composition. *Gastroenterology*. 2010; **139**: 1942-51 e2.

[12] Chaar V, Keclard L, Diara JP*, et al.* Association of UGT1A1 polymorphism with prevalence and age at onset of cholelithiasis in sickle cell anemia. *Haematologica*. 2005; **90**: 188-99.

[13] Passon RG, Howard TA, Zimmerman SA, Schultz WH, Ware RE. Influence of bilirubin uridine diphosphate-glucuronosyltransferase 1A promoter polymorphisms on serum bilirubin levels and cholelithiasis in children with sickle cell anemia. *J Pediatr Hematol Oncol*. 2001; **23**: 448-51.

[14] Hu ZY, Yu Q, Zhao YS. Dose-dependent association between UGT1A1*28 polymorphism and irinotecan-induced diarrhoea: a meta-

This article is protected by copyright. All rights reserved.

Accepted Article

analysis. *Eur J Cancer*. 2010; **46**: 1856-65.

[15] Ehmer U, Kalthoff S, Fakundiny B*, et al.* Gilbert syndrome redefined: a complex genetic haplotype influences the regulation of glucuronidation. *Hepatology*. 2012; **55**: 1912-21.

[16] Stocker R, Yamamoto Y, McDonagh AF, Glazer AN, Ames BN. Bilirubin is an antioxidant of possible physiological importance. *Science*. 1987; **235**: 1043-6.

[17] Lin JP, O'Donnell CJ, Schwaiger JP*, et al.* Association between the UGT1A1*28 allele, bilirubin levels, and coronary heart disease in the Framingham Heart Study. *Circulation*. 2006; **114**: 1476-81.

[18] Yesilova Z, Serdar M, Ercin CN*, et al.* Decreased oxidation susceptibility of plasma low density lipoproteins in patients with Gilbert's syndrome. *Journal of gastroenterology and hepatology*. 2008; **23**: 1556-60.

[19] Chen YH, Hung SC, Tarng DC. Serum bilirubin links UGT1A1*28 polymorphism and predicts long-term cardiovascular events and mortality in chronic hemodialysis patients. *Clin J Am Soc Nephrol*. 2011; **6**: 567-74.

[20] Lin R, Wang Y, Fu W*, et al.* Common variants of four bilirubin metabolism genes and their association with serum bilirubin and coronary artery disease in Chinese Han population. *Pharmacogenetics and genomics*. 2009; **19**: 310-8.

[21] Booth N. What are the Read Codes? *Health Libr Rev*. 1994; **11**: 177-

This article is protected by copyright. All rights reserved.

Accepted Article

82.

[22] Bourke A, Dattani H, Robinson M. Feasibility study and methodology to create a quality-evaluated database of primary care data. *Inform Prim Care*. 2004; **12**: 171-7.

[23] Blak BT, Thompson M, Dattani H, Bourke A. Generalisability of The Health Improvement Network (THIN) database: demographics, chronic disease prevalence and mortality rates. *Informatics in primary care*. 2011; **19**: 251-5.

[24] Khan NF, Harrison SE, Rose PW. Validity of diagnostic coding within the General Practice Research Database: a systematic review. *Br J Gen Pract*. 2010; **60**: e128-36.

[25] Maguire A, Blak BT, Thompson M. The importance of defining periods of complete mortality reporting for research using automated data from primary care. *Pharmacoepidemiol Drug Saf*. 2009; **18**: 76-83.

[26] Dave S, Petersen I. Creating medical and drug code lists to identify cases in primary care databases. *Pharmacoepidemiol Drug Saf*. 2009; **18**: 704-7.

[27] Sedlak TW, Saleh M, Higginson DS, Paul BD, Juluri KR, Snyder SH. Bilirubin and glutathione have complementary antioxidant and cytoprotective roles. *Proc Natl Acad Sci U S A*. 2009; **106**: 5171-6.

[28] Sedlak TW, Snyder SH. Bilirubin benefits: cellular protection by a

This article is protected by copyright. All rights reserved.

Accepted Article

biliverdin reductase antioxidant cycle. *Pediatrics*. 2004; **113**: 1776-82.

[29] Dennery PA, McDonagh AF, Spitz DR, Rodgers PA, Stevenson DK. Hyperbilirubinemia results in reduced oxidative injury in neonatal Gunn rats exposed to hyperoxia. *Free Radic Biol Med*. 1995; **19**: 395-404.

[30] Frei B, Stocker R, Ames BN. Antioxidant defenses and lipid peroxidation in human blood plasma. *Proc Natl Acad Sci U S A*. 1988; **85**: 9748-52.

[31] Neuzil J, Stocker R. Free and albumin-bound bilirubin are efficient co-antioxidants for alpha-tocopherol, inhibiting plasma and low density lipoprotein lipid peroxidation. *J Biol Chem*. 1994; **269**: 16712-9.

[32] Stender S, Frikke-Schmidt R, Nordestgaard BG, Grande P, Tybjaerg-Hansen A. Genetically elevated bilirubin and risk of ischaemic heart disease: three Mendelian randomization studies and a meta-analysis. *Journal of internal medicine*. 2013; **273**: 59-68.

[33] Najib F. Defensive role of bilirubinemia in pneumococcal infection. *The Lancet*. 1937; **229**: 505-6.

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Table 1: Baseline characteristics of a cohort with recognised Gilbert's syndrome and a comparison cohort selected to have similar age and gender distribution but bilirubin levels ≤17 µmol/L.

	Gilbert's cohort (n=4,266)	Comparison cohort (n=21,968)	Univariable analysis †
Male (%)	2,871 (67)	14,711 (67)	NS
	Median (IQR)	Median (IQR)	
Serum total bilirubin µmol/L	29 (24-37)	10 (7-12)	P<0.001
Follow-up in years	9 (5-12)	9 (5-12)	NS
Age at bilirubin test	48 (35-61)	49 (36-61)	P<0.001
BMI	24.9 (22.3-27.8)	26.0 (23.2-29.5)	P<0.001
DBP, mmHg	80 (70-86)	80 (71-88)	P<0.001
SBP, mmHg	130 (120-143)	130 (120-145)	P<0.001
ALP, IU/L	71 (57-91)	76 (61-99)	P<0.001
ALT, IU/L	24 (17-33)	25 (18-36)	P<0.001
Smoking status (%)			
Current	427 (10)	6,250 (28)	
Ex	800 (19)	4,252 (19)	
Never	3,039(71)	11,466 (52)	P<0.001
Social deprivation score (%)			
1 (least deprived)	1,402 (33)	6,192 (28)	
2	1,104 (26)	5,132 (23)	
3	856 (20)	4,671 (21)	
4	622 (15)	3,795 (17)	
5 (most deprived)	282 (7)	2,178 (10)	P<0.001
Co-morbidity (%)			
CVD	171 (4)	1,488 (7)	0.04
Type II diabetes	90 (2)	1,405 (6)	P<0.001
Chronic respiratory disease	115 (3)	1,210 (6)	P<0.001
Mental illness	498 (12)	5,326 (24)	

Abbreviations: ALP, alkaline phosphatase; ALT, alanine aminotransferase ; BMI, body mass

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index; CVD, cardiovascular disease, DBP, diastolic blood pressure; NS, not statistically significant (P>0.05); SBP, systolic blood pressure; SD, standard deviation.

† P-values are from univariable linear regression for continuous variables after transformation if data are skewed. Significant of categorical variables were assessed using a Wald test.

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Table 2: Mortality rates in a cohort of patients with Gilbert's syndrome and a comparison cohort without evidence of the condition.

Gilbert's cohort		Comparison cohort		Model 1 †		Model 2 ‡	
Events/PYs (per 10,000)	Mortality rate (95%CI)	Events/PYs (per 10,000)	Mortality rate (95%CI)	MRR (95%CI)	p-value	MRR (95%CI)	p-value
97/4.05	23.9 (19.6-29.2)	1077/21.6	49.8 (46.9-52.9)	0.43 (0.34-0.55)	<0.001	0.53 (0.40-0.66)	<0.001

Abbreviations: CI, confidence interval; MRR, mortality rate ratio; PYs, person years

† Model 1=age, gender, time period

‡ Model 2=multivariable analysis adjusting for age, gender, year of cohort entry, smoking status, blood pressure, body mass index, alanine aminotransferase, alkaline phosphatase, and social deprivation index.

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